Small Object Detection in Remote Sensing Images Based on Redundant Feature Removal and Progressive Regression

Yang Yang, *Member, IEEE*, Bingjie Zang, Chunying Song, Beichen Li, Yue Lang, *Member, IEEE*, Wenyuan Zhang, and Peng Huo

***Abstract*— Small object detection in large-scale remote sensing images (RSIs) is crucial for military and civil applications, but it remains challenging. Since small objects occupy few pixels, their features are easily interfered with by complex backgrounds and large objects. In addition, they are susceptible to localization offsets, which are prone to false or missed detections as there are few predicted bounding boxes matching the ground truth. To overcome these issues, this article proposes a filter progressive small object detection (FPSOD) model that is based on the progressive mechanism. With the proposed attention-based soft-threshold filtering module, FPSOD significantly filters out redundant information in high-level feature maps thus enhancing the semantic features of small objects. Furthermore, a progressive regression loss (PR-Loss) function is proposed to facilitate the precise localization, which mitigates predicted bounding box drift by limiting the fluctuated range of the gradients. The experimental results show that the proposed model substantially improves the precision and recall of small objects, effectively reduces missed detections, and improves detection performance.**

***Index Terms*— Feature filter, progressive regression, remote sensing images (RSIs), small object detection.**

I. Introduction

THE advent of remote sensing imaging technology has garnered significant interest in the detection of objects in remote sensing images (RSIs) for purposes such as disaster assessment [1], military object identification [2], and intelligent security [3]. RSIs are commonly acquired using devices like drones [4] and satellites [5], which capture comprehensive visual data of specific geographic regions from the look-down view. Consequently, it can be observed that common objects found in real life, such as vehicles, ships [6], [7], and airplanes, are represented by a limited number of pixels within the RSIs, leading to a dearth of extensive information pertaining to these small objects [8], [9]. As a result, small object detection continues to pose a challenge in RSIs [10], as shown in Fig. 1.

In order to enhance the precision of detecting objects, researchers have conducted thorough investigations on multiscale feature extraction techniques. One of the prominent examples is the feature pyramid network (FPN) [11], which integrates feature maps from various stages in a hierarchical manner to propagate semantic information. Given that the absolute resolutions (lower than 32×32 [12]) or related scales associated with the panoramic view of the RSI (no more than 0.12% of the RSI [13]) are utilized to describe the small objects, the primary concern regarding the detection model becomes the confirmation of the object's existence. In small object detection, researchers typically enhance the feature extraction capability of FPN from both scale and space perspectives, proposing variant models like TDFP [14], scale sequence (S^2) feature [15], and EFPN [16]. The features extracted by these models are typically scale-invariant and space-invariant, reducing the risk of missed detections when detecting small objects at different FPN layers. However, such models remain susceptible to the interference of redundant features. Specifically, the features of small objects extracted by the backbone network are prone to interference from background information, as semantic information of them becomes increasingly difficult to capture with the continuous downsampling of the convolutional neural network [17]. This interference effect gradually increases with the upsampling of the feature map in the FPN-based neck network, posing greater challenges for small object detection.

Upon extracting and fusing multiscale features via the neck network, the detection head network is still required to execute recognition and localization. The current method of determining the precise location entails utilizing the intersection over union (IoU) metric, which quantifies the overlap between the predicted bounding box and the ground-truth bounding box. However, the variability of IoU sensitivity in relation to objects of varying scales is significant, and even minor offset of the bounding boxes can significantly disturb the IoU values for small objects [18]. Consequently, the predicted bounding box, which is regarded as a positive sample in the present training

Manuscript received 16 June 2023; revised 5 December 2023 and 2 May 2024; accepted 15 June 2024. Date of publication 24 June 2024; date of current version 5 July 2024. This work was supported in part by the Digital Grid Open Fund of China Southern Power Grid Company Ltd., in part by the National Key Research and Development Program of China under Grant 2021YFE0204200, and in part by the National Natural Science Foundation of China under Grant 62101378 and Grant 62171318. *(Corresponding author: Chunying Song.)*

Yang Yang, Bingjie Zang, Chunying Song, and Beichen Li are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: scy@tju.edu.cn).

Yue Lang is with the School of Electric and Information Engineering, Hebei University of Technology, Tianjin 300401, China.

Wenyuan Zhang is with the School of Environment and Spatial Informatics, China University of Mining and Technology, Xuzhou 221116, China.

Peng Huo is with the High Voltage System Department, Longyuan (Beijing) Renewable Energy Engineering Technology Company Ltd., Beijing 100034, China.

Digital Object Identifier 10.1109/TGRS.2024.3417960

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

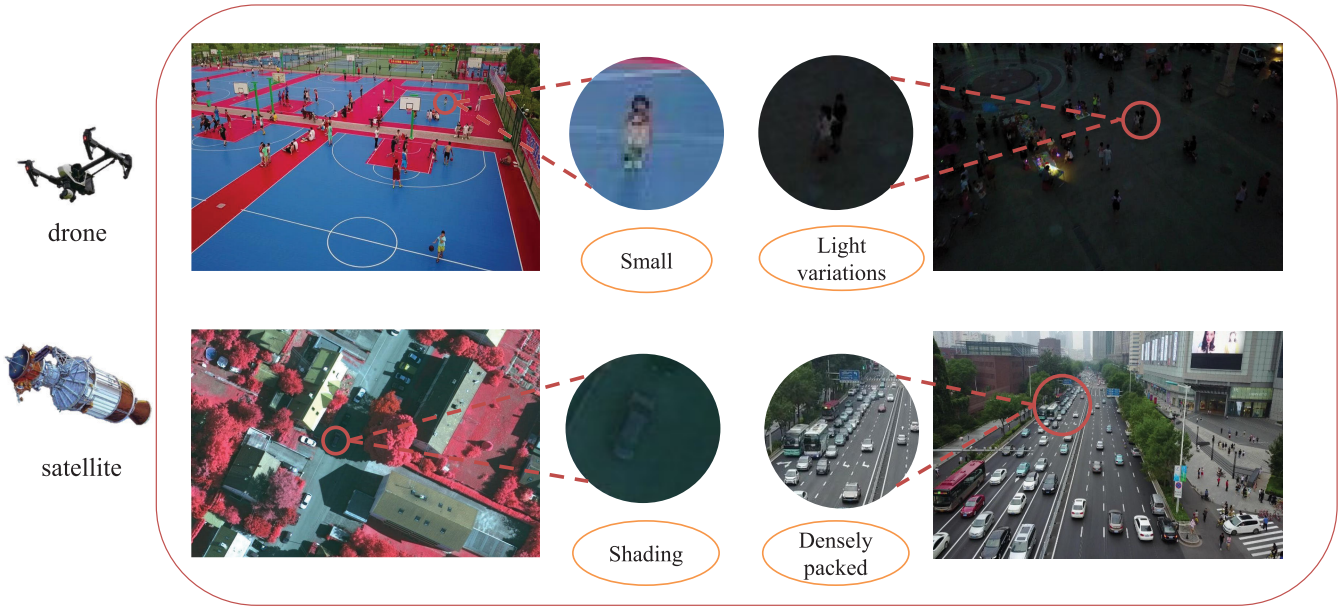


Fig. 1. Challenges of small-scale remote sensing object detection are influenced by various factors, including resolution, illumination, shadows, and object density.

epoch, deviates from the ground truth owing to significant gradients in the subsequent epochs, leading to severe missing detections. As a consequence, the key of small object detection is to avoid missing small objects resulting from their sensitivity to offset.

In this article, we propose a progressive one-stage object detector for small object detection in RSIs. To alleviate the issue of inaccurate semantics of small objects in high-resolution feature maps caused by the amplification of redundant features in the FPN, we develop a feature filtering pyramid approach that enhances the semantic information of small objects in high-resolution feature maps by filtering out irrelevant information. Furthermore, to tackle the problem of the sensitivity of the IoU metric to the relative position offset between predicted bounding boxes and ground truth for small objects, we introduce a progressive regression loss (PR-Loss), which aims to gradually bring positive samples with low IoU closer to the ground truth.

The main contributions of this article are summarized as follows.

1) We propose a one-stage detector based on the progressive mechanism, named filter progressive small object detection (FPSOD), which effectively addresses the issue of missed small object recognition in remote sensing scenes that current detectors face. Experimental results on the large-scale and multiscene dataset Visdrone and optical remote sensing dataset NWPU VHR-10 indicate that our detector outperforms all compared models and achieves 15.1%–34.4% in mean average precision on small objects (mAP@$_S$), respectively.
2) We construct an improved FPN using the shrinkage attention mechanism. It aims to improve the semantic information proportion of small objects in the upsampling phase of the Neck network, effectively mitigating the interference of redundant features on the detection of small objects in high-resolution feature maps.
3) We propose a universal PR-Loss function based on the raised cosine function. It implements IoU-based adaptive weighting and reduces the update rate of positive anchors with low IoU, thus reducing the sensitivity of the IoU metric to the relative position offset. Additionally, we validate the effectiveness of this formula on various loss functions.

The remaining sections are organized as follows. Section II reviews related works, while Section III describes the details of the proposed model. In Section IV, we provide implementation details and experimental results. Finally, the conclusion is summarized in Section V.

II. Related Works

A. Small Object Detection in RSIs

In recent years, significant progress has been made in the field of object detection through the use of deep learning methods. These methods rely on deep neural networks to extract features from RSIs and predict object locations and categories using either a one-stage or two-stage method. However, despite advancements, these detectors continue to face challenges in accurately detecting objects across different scales. In the RSIs, small objects tend to occupy only a limited portion of the overall image, which consequently leads to a scarcity of available samples for training. To address this limitation, researchers have delved into exploring data augmentation techniques that involve segmentation and resizing [19], to expand the pool of training samples for small objects. On the other hand, small objects inherently suffer from lower resolution and limited information, which can potentially result in inferior representations. To address this challenge, Xiaolin et al. [20] proposed a specialized super-resolution (SR) module designed to enhance the extraction of features for small objects. This module enables the detection of these objects within the low-level feature maps. Bashir and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

Wang [21] advanced the existing SR framework by integrating generative adversarial networks (GANs) and residual feature aggregation techniques. This integration enables the capture of complex features in the SR process. Moreover, researchers have employed contextual cues to enhance the detection of small objects. FS-SSD [22] enhances detection accuracy by integrating the spatial contextual relationships among multiple object classes within a specific distance into the object re-detection process. However, these methods either compromise model interpretability or escalate model complexity, with GAN-based approaches, in particular, introducing additional complexities to the optimization process. Nevertheless, the key to improving detection accuracy lies in effectively extracting relevant features and achieving precise localization of small objects.

B. Multiscale Feature Representation

Multiscale feature fusion can improve detection performance and robustness for objects of different sizes. The FPN [11] builds high-level semantic representations for objects of all sizes by obtaining feature maps of various perceptual fields from the backbone network and constructing feature pyramids from the top down. Unlike FPN, which utilizes the unidirectional fusion structure, PANet [23] proposed a bidirectional fusion network with top-down and bottom-up paths. The latter shortens the information connection between the bottom and top layers, propagates more robust semantic information from the lower layers, and improves the localization capability of the feature layers.

Subsequently, Tan et al. [24] discovered that feature layers with different resolutions contribute unequally to fusion, and proposed a bidirectional fusion feature pyramid structure with adaptive weights named BiFPN. By learning useful information on different feature layers via adaptive weights, BiFPN enables simpler and faster information fusion. Qiao et al. [25] explored the performance of the mechanism of looking and thinking twice and proposed the Recursive-FPN. The Recursive-FPN connects the additional feedback information from FPN to the bottom-up backbone network via atrous spatial pyramid pooling module and recursively improves the feature pyramid to generate a more robust feature representation. Furthermore, Liu et al. [26] explored the challenge of insufficient utilization of scene semantic information between different regions in the feature pyramid by proposing a dual relationship module that learns multiscale latent relationships based on dual relationships. The bidirectional module generates dynamic weights to facilitate the fusion of features at different scales, thereby enhancing the multiscale feature representation. While the above methods have shown good performance, they fail to consider the impact of redundant features on the semantic representation of objects, which particularly causes significant interference in small object detection.

C. Localization Loss Function

Supervised object detectors use ground-truth labels to constrain the location of anchors. Early object detection models, such as Faster RCNN [27] and SSD [28], the localization loss is calculated using $L1$ loss and its improved forms, including Smooth $L1$ Loss [29] and Balanced $L1$ Loss [30]. These loss functions evaluate the overlap between predicted and ground-truth bounding boxes based on differences in their coordinates. In recent research, IoU has attracted considerable attention due to its scale invariance, which renders it better suited than $L1$ loss for measuring the overlap between predicted and ground-truth bounding boxes.

IoULoss [31] is an early application of IoU regression bounding boxes. Rather than separately regressing the prediction box coordinates using $L1$ Loss and $L2$ Loss, IoULoss considers the prediction box as a single entity. Due to its advantageous characteristics, IoULoss has been highly effective and is commonly utilized in various object detection models.

However, when the predicted bounding box fails to intersect with the ground truth, IoULoss can cause the issue of gradient vanishing. In this regard, Rezatofighi et al. [32] introduced GIoULoss to address this issue by introducing a penalty term. This penalty is determined by the proportion of empty space in the smallest area calculated by the predicted bounding box and the ground truth. Zheng et al. [33] suggested that a better localization regression should consider several factors, including the overlap area, central point distance, and aspect ratio. In response, they developed DIoULoss and CIoULoss [33], which converge faster and perform better. Zhang et al. [34] focused on the effective example mining problem in anchor regression and proposed the Focal-EIoU loss to balance high and low-quality anchors during training. He et al. [35] extended the existing linear IoU-based loss with Box-Cox transformation, yielding a more robust loss for noisy data. However, these localization loss functions do not take into account the sensitivity of small objects to predicted bounding box offsets, which may result in the missed detections.

III. Methodology

This section offers a comprehensive portrait of our proposed FPSOD model. We first present the overall architecture of the model, followed by delineation of the interference removal module and PR-Loss individually. Lastly, the loss functions utilized within the proposed model are summarized.

A. Overview

The overall architecture of the proposed model is depicted in Fig. 2. In general, the model comprises three parts: the backbone, soft-thresholding filter FPN, and the detection head.

The ResNet50 [36] is utilized as the backbone network for feature extraction, consisting of five stages. Stage 0 is dedicated to image preprocessing, while stages 1 through 4 are dedicated to feature extraction, with 3, 4, 6, and 3 bottleneck modules, respectively. The progressive reduction of feature map size from stages 0 to 4 leads to a gradual improvement in semantic information. The outputs of stage 2∼4 in the backbone are denoted as $C3$, $C4$, and $C5$, and top-down connections are established by them.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

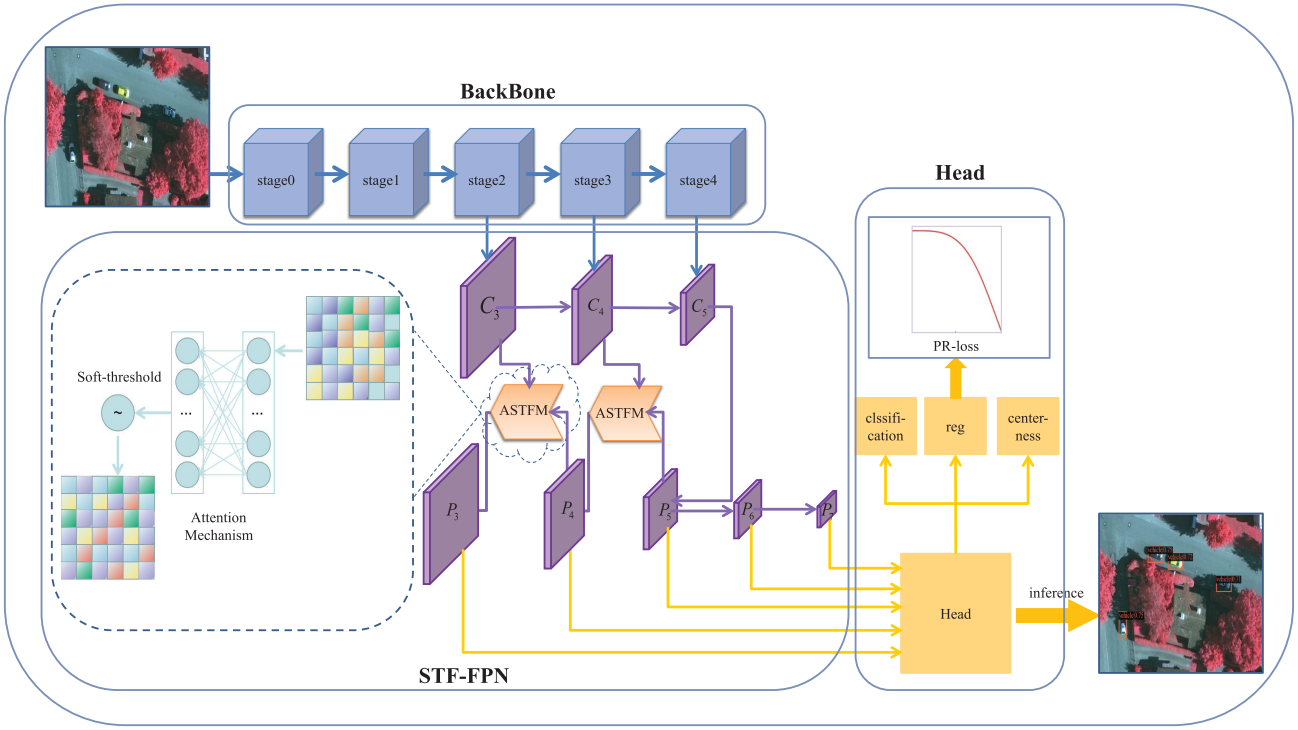


Fig. 2. Overall architecture of the proposed model.

The soft-thresholding filtering FPN (STF-FPN) is built upon the basis of FPN [11] and incorporates a top-down connection to facilitate the learning of objects at multiple scales. The central component of it, the attention-based soft-thresholding filter module (ASTFM), is positioned between every two upsampling layers of FPN. It leverages soft thresholding and attention mechanisms to effectively mitigate the issue of the interference amplification of redundant features that arises as a result of the upsampling process. The ultimate segment of the network comprises a detection head module that encompasses three branches, namely classification, localization, and centerness. To optimize the localization of small objects, we propose the PR-Loss.

During the inference phase, the detection head network predicts the object's location and eliminates the redundant boxes through the non-maximum suppression algorithm to generate the final detection results.

B. Soft-Thresholding Filtering Feature Pyramid Network

During the process of feature extraction, redundant features such as background information could be included, which can interfere with the semantic expression of small objects and lead to missing detections [17]. This motivates us to design a feature filter within the FPN to filter out interference information layer by layer from top to bottom, thereby enhancing the semantic information of small objects while maintaining the semantics of large ones. Therefore, we propose the ASTFM and place it between the upsampling layers of FPN to achieve redundant feature filtering. The module is drawn in Fig. 3. It mainly consists of the following four parts: feature pooling, convolution operation, soft-threshold generation, and redundant feature filtering.

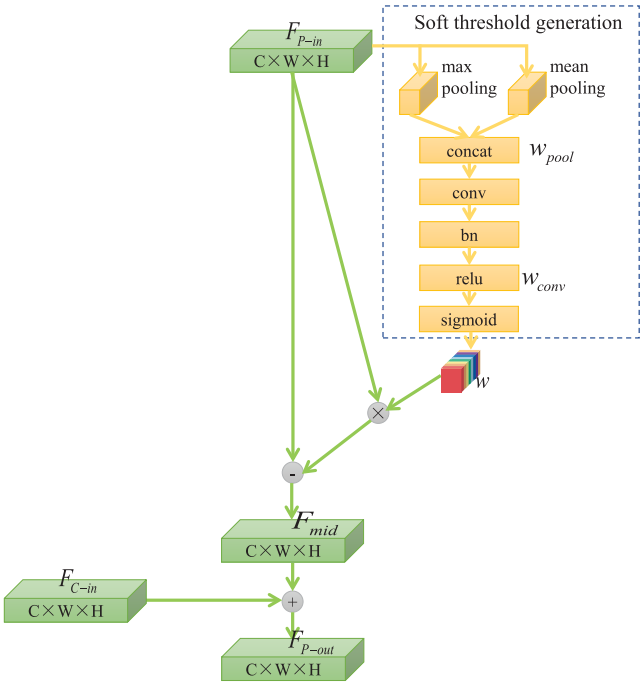


Fig. 3. Soft-threshold redundant removal module.

Step 1 (Feature pooling): The input of the module is the layer of the feature pyramid, denoted as $F_{P-\text{in}} \in \mathbb{R}^{C\times H\times W}$. The implementation of max pooling and average pooling operations is then employed to compress the channels of feature maps, thereby producing two pooling layers P_1, $P_2 \in \mathbb{R}^{1\times H\times W}$. These two pooling layers are concatenated in channel dimension, and $\omega_{\text{pool}} \in \mathbb{R}^{2\times H\times W}$ is generated. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

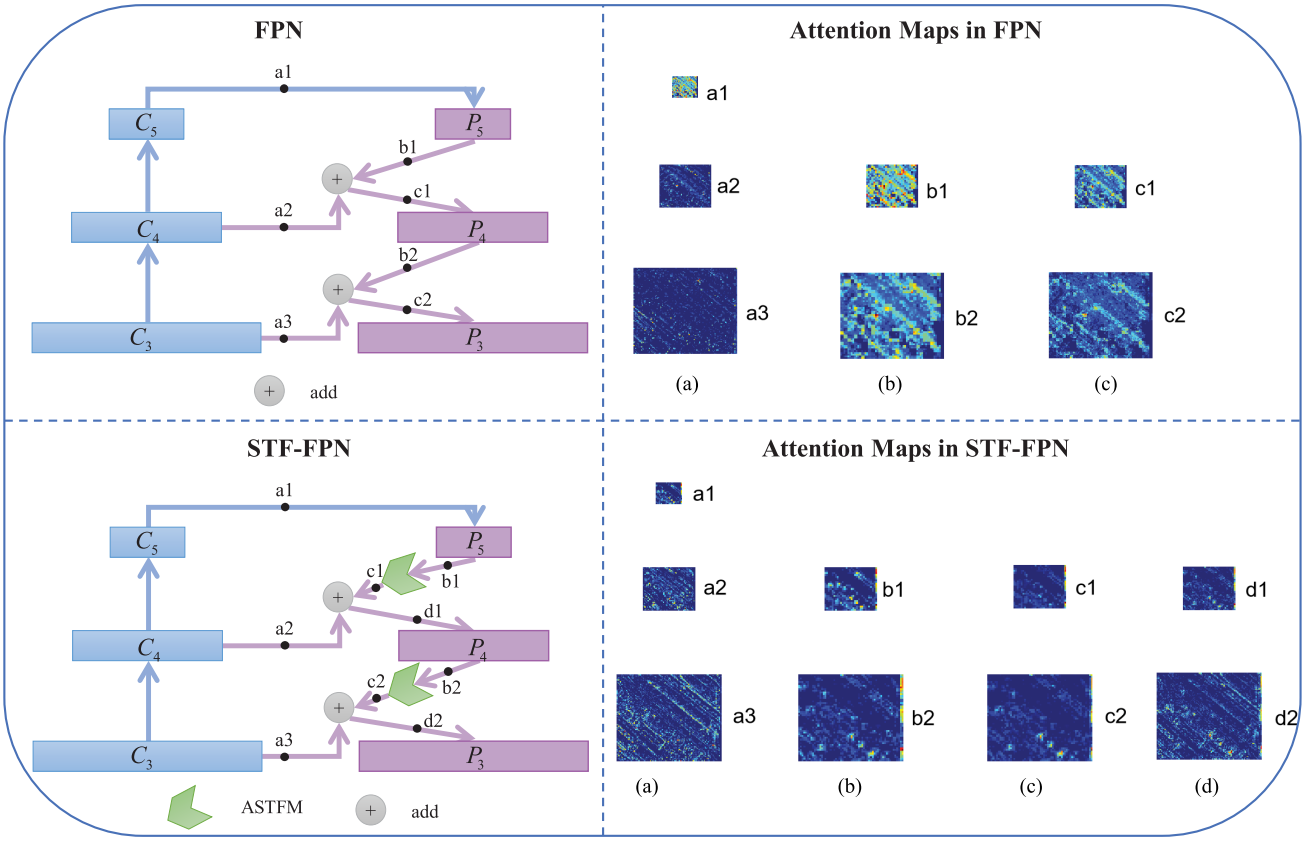


Fig. 4. Comparison of FPN and STF-FPN.

above process can be expressed as the following equation:

$$\omega_{\text{pool}} = \text{concat}(\text{MaxPooling}(F_{P-\text{in}}), \text{AvgPooling}(F_{P-\text{in}})). \quad (1)$$

Step 2 (Convolution operation): After feature pooling, we feed the ω_{pool} to the convolutional layer and the output is normalized by the batch normalization. In addition, we employ the rectified-linear-unit (ReLU) activation function to increase the nonlinear expressiveness of the pooling layer and generate ω_{conv}

$$\omega_{\text{conv}} = \text{ReLU}\big(\text{BatchNormal}\big(\text{Conv}\big(\omega_{\text{pool}}\big)\big)\big). \quad (2)$$

Step 3 (Soft-threshold generation): The sigmoid activation function maps ω_{conv} to the range of 0–1, obtaining the soft threshold of the redundant input feature map at each position, which is expressed as follows:

$$w = \text{sigmoid}(w_{\text{conv}}) = \frac{1}{1 + e^{-w_{\text{conv}}}}. \quad (3)$$

Step 4 (Redundant feature filtering): The redundant features are eliminated by filtering the input feature layer with the soft threshold generated in step 3. Specifically, this is achieved by subtracting the redundant feature information from the input feature layer

$$F_{\text{mid}} = F_{P-\text{in}} - F_{P-\text{in}} \times w. \quad (4)$$

Finally, the feature layer from the bottom-up pathway is element-wise added to the filtered feature layer

$$F_{P-\text{out}} = F_{C-\text{in}} + F_{\text{mid}}. \quad (5)$$

In order to attain progressive redundant elimination, the ASTFM module is inserted multiple times in the FPN instead of using it only once, leading to the development of STF-FPN as shown in Fig. 4. From the figure, it can be seen that the information in the FPN feature maps is cluttered and redundant, while the proposed STF-FPN can represent the features of the objects effectively.

C. Progressive Regression IoULoss

1) Small Object's Susceptibility to Localization Offset: As mentioned in the introduction, small objects are highly susceptible to localization offset, as illustrated in Fig. 5. In the given figure, A and B are two predictions while GT denotes the ground truth of the object. It can be obviously noted that, when the object is 6 × 6 resolution, the reduction in IoU is less than the counterparts regarding the 3 × 3 object.

The offset-IoU curves of objects with different resolutions (3 × 3 and 6 × 6 resolution for the red and blue lines, respectively) are depicted in Fig. 6. The figure provides evidence that smaller objects are more susceptible to localization offset. The X-axis in Fig. 6 denotes the differences in position between the predicted bounding box and the actual ground truth, while the Y-axis denotes the IoU value between the two entities.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

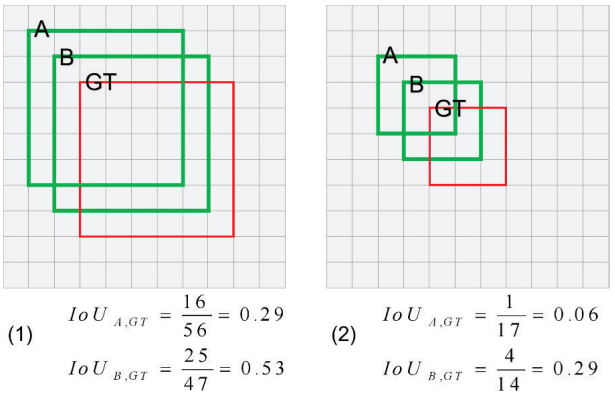


Fig. 5. Influence of prediction box deviates on objects of different scales.

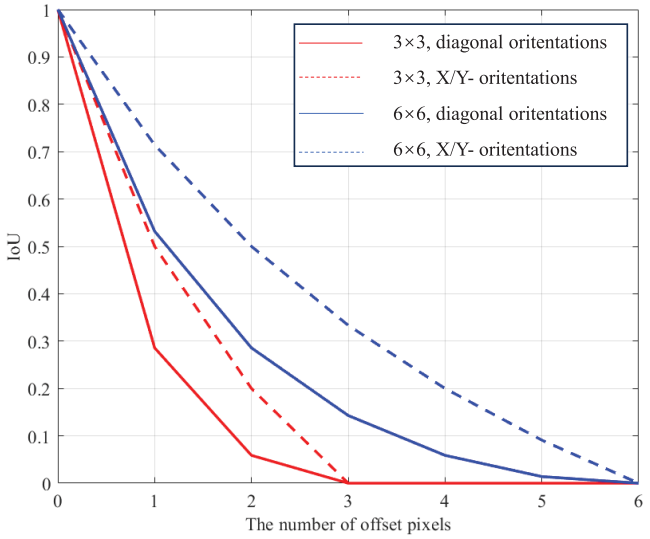


Fig. 6. Offset-IoU curve.

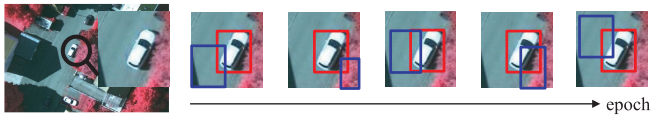


Fig. 7. Prediction bounding box drift caused by large gradients.

The solid line represents the presence of offset in diagonal orientations (as shown in Fig. 5), whereas the dashed line represents offset in either the X- or Y-orientations.

2) Bounding Box Drifts in Small Object Detection: This susceptibility issue raises a challenge in precisely localization to the small object, in particular the predicted box would drift during training (see Fig. 7).

In the early phases of training, it is common for the localization error, as determined by the regression loss function, to be relatively high. As the training progresses and converges, the error gradually diminishes. Consequently, the predicted bounding box aligns more closely with the ground-truth representation of the object. In contrast, the susceptibility to the localization offset results in a persistent high regression loss value when the predicted box is in close proximity to the ground truth, particularly when dealing with small objects. The occurrence described results in a significant gradient (the difference in gradient between IoU loss and the proposed loss is presented in the following subsection) that is backpropagated to the neural network, thereby causing unpredictability in the localization during the subsequent epoch. The drifting of the bounding box occurs as a consequence.

3) Progressive Regression IoULoss: The objective of our work is to reduce the influence on the box drifts with respect to the small objects. In this section, we will introduce the proposed PR-Loss, and this loss function is generalized based on the raised cosine function, which can be represented by

$$f(x) = C\left(\alpha + \lambda \cos\left(\frac{(x^n - \mu)\pi}{\sigma}\right)\right) \tag{6}$$

where C, α, λ, μ, σ, and n are the parameters for describing the shape of this function, and specifically the C, λ, σ, and n are the scale factors, α and μ denote the offset. By utilizing these specified parameters, the loss function offers significant adaptability in preventing bounding box drift. The subsequent sections will demonstrate how these parameters impact the IoU-loss curves. For the sake of convenience in our research, we set μ to 0 in our research.

Upon replacing the x with IoU, the PR-Loss can be formulated as follows:

$$L_{\text{PR}-\text{IoU}}(\text{IoU}) = C\left(\alpha + \lambda \cos\left(\frac{(\text{IoU}^n)\pi}{\sigma}\right)\right), \quad \text{IoU} \in [0, 1]. \tag{7}$$

In order to align the loss values with the boundaries of the cosine function, two constraints are initially employed to limit the upper and lower bounds of the loss function. These constraints are defined as $L_{\text{PR}-\text{IoU}}(0) = 1$ and $L_{\text{PR}-\text{IoU}}(1) = 0$. In addition, we also guarantee the monotonically decreasing property of this function by enforcing $\nabla_{\text{IoU}} L_{\text{PR}-\text{IoU}} \leq 0$, which ensures that a low IoU value corresponds to a large loss.

Moreover, the proposed function form can be integrated with other loss functions, such as GIoU [32], DIoU, and CIoU [33]. Our PR-Loss can be combined with these methods to formulate a comprehensive loss function, which can be expressed as follows:

$$\begin{aligned} L_{\text{GIOU}} &= 1 - \text{IoU} + \frac{\left|D \backslash \left(B \cup B^{\text{gt}}\right)\right|}{|D|} \Rightarrow L_{\text{PR}-\text{GIoU}} \\ &= C\alpha + C\lambda \cos\left(\frac{\pi}{\sigma}\left(\text{IoU} - \frac{\left|D \backslash \left(B \cup B^{\text{gt}}\right)\right|}{|D|}\right)^n\right) \end{aligned} \tag{8}$$

$$\begin{aligned} L_{\text{DIOU}} &= 1 - \text{IoU} + \frac{\rho^2\left(\mathbf{b}, \mathbf{b}^{\text{gt}}\right)}{d^2} \Rightarrow L_{\text{PR}-\text{DIOU}} \\ &= C\alpha + C\lambda \cos\left(\frac{\pi}{\sigma}\left(\text{IoU} - \frac{\rho^2\left(b, b^{\text{gt}}\right)}{d^2}\right)^n\right) \end{aligned} \tag{9}$$

$$\begin{aligned} L_{\text{CIoU}} &= 1 - \text{IoU} + \frac{\rho^2\left(\mathbf{b}, \mathbf{b}^{\text{gt}}\right)}{d^2} + \alpha\upsilon \Rightarrow L_{\text{PR}-\text{CIoU}} \\ &= C\alpha + C\lambda \cos\left(\frac{\pi}{\sigma}\left(\text{IoU} - \frac{\rho^2\left(b, b^{\text{gt}}\right)}{d^2} - \alpha\upsilon\right)^n\right) \end{aligned} \tag{10}$$

where D in L_{GIoU} means the smallest enclosing convex object for bounding box B and the ground truth B^{gt}, and C means the diagonal length of D. The term $\rho^2(\mathbf{b}, \mathbf{b}^{\text{gt}})$ in L_{DIoU} denotes the squared Euclidean distance between the central points $\mathbf{b}$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

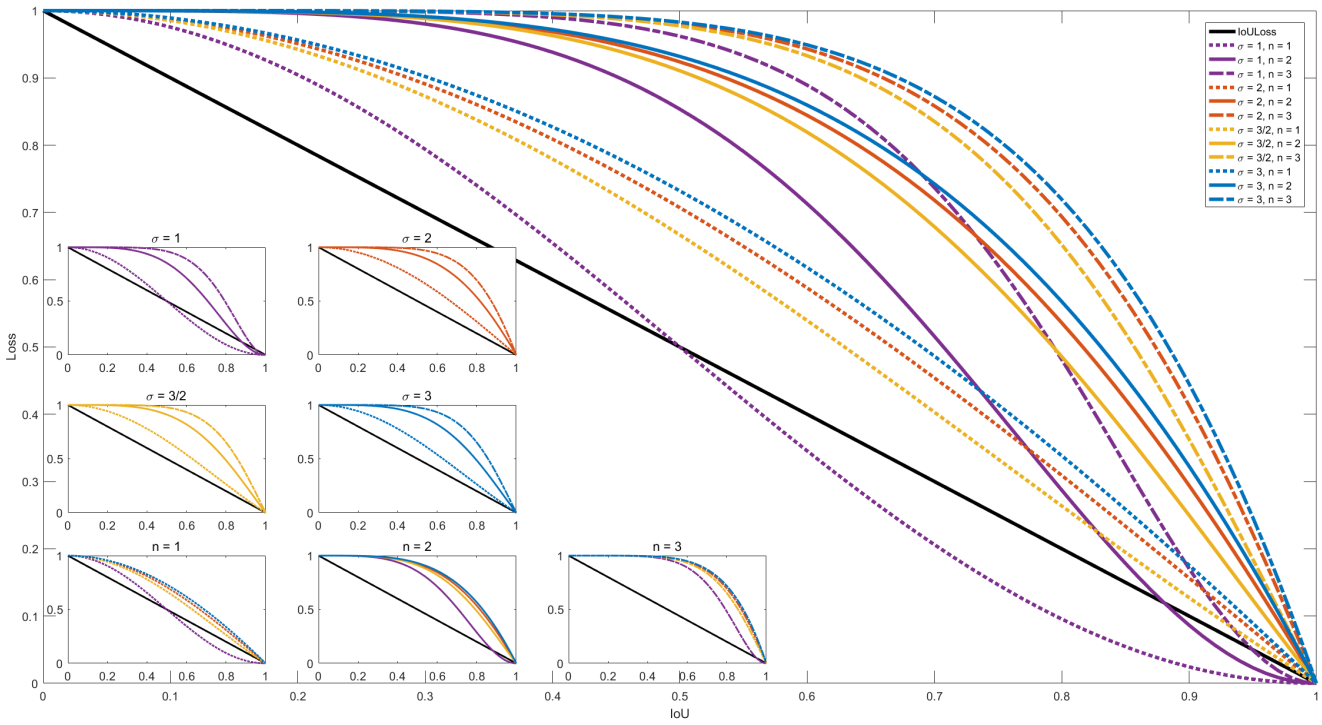


Fig. 8. Correlation between IoU and PR-IoULoss with different σ and n.

and $\mathbf{b}^{\mathrm{gt}}$ of B and B^{gt}, respectively. α in L_{CIoU} is a parameter for trade-off, and υ is a measurement parameter for the consistency of aspect ratio.

4) Properties of PR-Loss: In this part, we will take the PR-IoULoss as an example and discuss its properties.

Properity 1 (Loss Weighting): PR-Loss can be weighted to different IoUs in an adaptive manner. We draw the curves of PR-IoULoss in Fig. 8, where σ is 1, 2, 3/2, 3, and n is 1, 2, 3, respectively. From the figure, it can be observed that the PR-IoULoss weights the loss differently at various scales of IoU compared to the IoUloss.

Properity 2 (Gradient Weighting): The gradient of PR-Loss can be weighted to different IoUs in an adaptive manner.

Let $|\nabla_{\mathrm{IoU}} L_{\mathrm{PR-IoU}}|$ be the absolute gradient of $L_{\mathrm{PR-IoU}}$, $|\nabla_{\mathrm{IoU}} L_{\mathrm{IoU}}|$ be the absolute gradient of L_{IoU}, and ω_{grad} be the relative gradient weighting parameters between the two, which are defined as follows:

$$|\nabla_{\mathrm{IoU}} L_{\mathrm{PR-IoU}}| = C\lambda \frac{n\pi}{\sigma} \mathrm{IoU}^{n-1} \sin\left(\frac{\pi}{\sigma}\mathrm{IoU}^{n}\right) \tag{11}$$

$$\begin{aligned} \omega_{\mathrm{grad}} &= |\nabla_{\mathrm{IoU}} L_{\mathrm{PR-IOU}}| / |\nabla_{\mathrm{IoU}} L_{\mathrm{IoU}}| \\ &= C\lambda \frac{n\pi}{\sigma} x^{n-1} \sin\left(\frac{\pi}{\sigma}\mathrm{IoU}^{n}\right) / 1. \end{aligned} \tag{12}$$

Fig. 9 depicts the absolute gradient curves under different σ and n. In general, ω_{grad} is less than 1 when IoU is small and greater than 1 when IoU is large, which proves that the gradient of PR-Loss can be weighted to different IoUs in an adaptive manner. This encourages the network to progressively regress the object's localization during the initial phases of training, when the network's feature learning is not yet sufficient to achieve a high IoU for small objects. Consequently, the gradient of the loss function is small. As IoU increases, the network tends to learn object features in a more stable manner. A moderate increase in the loss gradient can help the network converge faster and accelerate its training speed.

D. Loss Functions

The loss functions employed in the FPSOD are formulated as follows:

$$L = L_{\mathrm{cls}} + L_{\mathrm{loc}} + L_{\mathrm{centerness}}. \tag{13}$$

1) Classification Loss L_{cls}: The classification loss is represented using the Focal loss, which is proposed in [37], and can be expressed as follows:

$$L_{\mathrm{cls}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{14}$$

$$p_t = \begin{cases} p, & \text{if } y = 1 \\ 1 - p, & \text{otherwise} \end{cases} \tag{15}$$

where the hyperparameters α_t and γ are set to 0.25 and 2, respectively.

2) Regression Loss L_{loc}: We employ the proposed $L_{\mathrm{PR-GIoU}}$ as the regression loss function and present it as follows:

$$L_{\mathrm{loc}} = C\alpha + C\lambda \cos\left(\frac{\pi}{\sigma}\left(\mathrm{IoU} - \frac{\left|D \backslash \left(B \cup B^{\mathrm{gt}}\right)\right|}{|D|}\right)^{n}\right) \tag{16}$$

where $C\alpha$ and $C\lambda$ are set to 1/3 and 2/3, respectively. σ is set to 3/2, and n is set to 2. The meaning of the other variables can be found above.

3) Centerness Loss $L_{centerness}$: Same with [38], we employ the centerness loss to suppress low-quality predicted bounding boxes. It can be represented as follows:

$$L_{\mathrm{centerness}} = -c_i \log c_i^{*} + (1 - c_i) \log\left(1 - c_i^{*}\right) \tag{17}$$

where c_i is the centerness of the predicted bounding box, and c_i^{*} is the ground truth.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

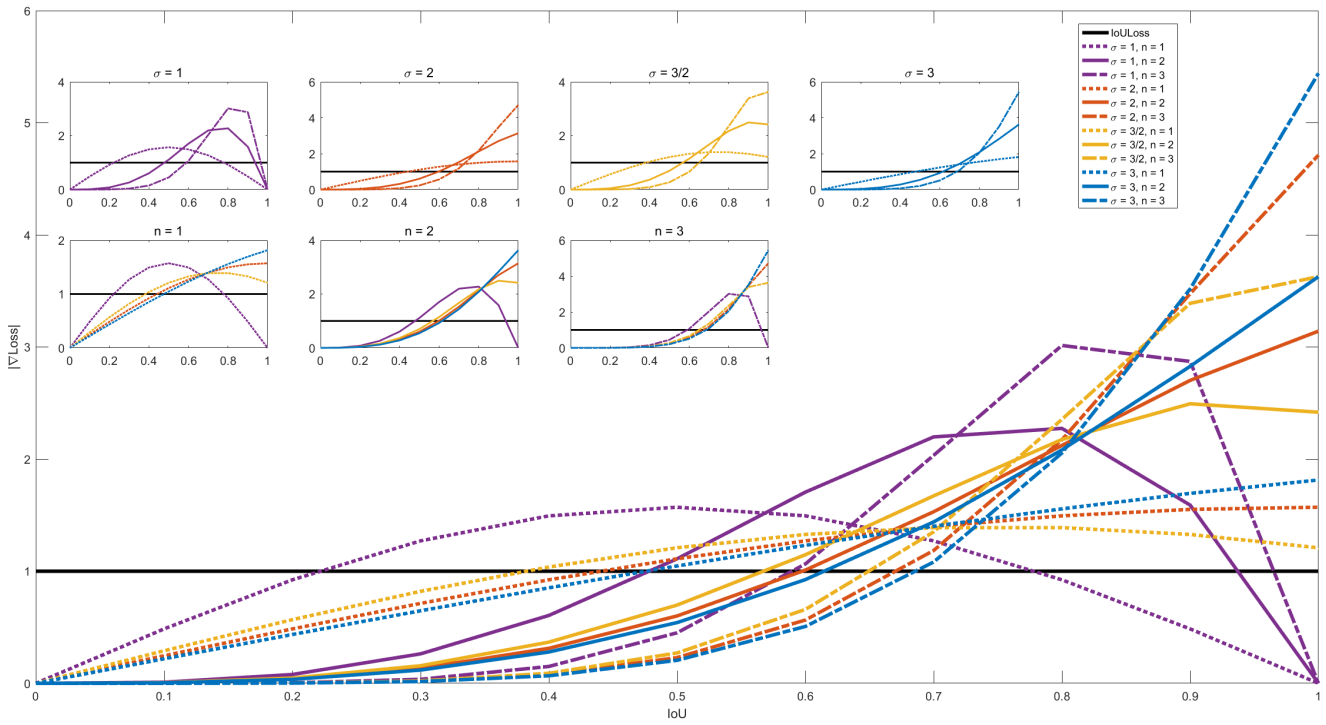


Fig. 9. Correlation between the absolute gradients of IoU and PR-IoULoss with different σ and n.

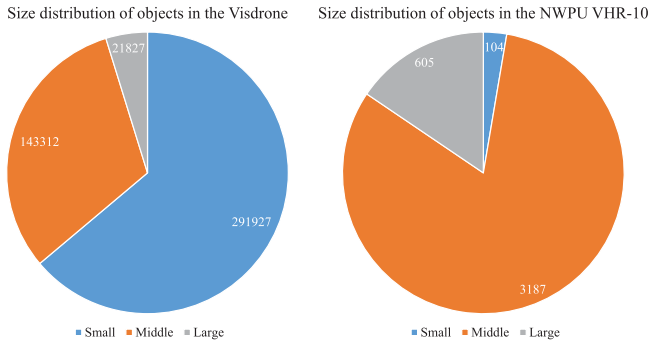


Fig. 10. Size distribution of objects in two datasets.

IV. Experiment

A. Datasets

Our experiments utilize two datasets: the Visdrone dataset [39] and the NWPU VHR-10 dataset [40], [41], [42]. The former is obtained by UAV and the latter by satellite.

1) Visdrone Dataset: This dataset is obtained by the AISKYEYE team using cameras mounted on different drone platforms, in diverse scenarios, and under varying weather and lighting conditions. It contains 12 distinct object types, of which we focused on ten categories by excluding the "ignored regions" and "other" categories in our experiments. The dataset is partitioned into three subsets, namely training set, validation set, and testing set, including 6471, 548, and 1610 images, respectively.

2) NWPU VHR-10 Dataset: The NWPU VHR-10 dataset is developed by Northwestern Polytechnical University. The dataset contains ten object classes with a total of 650 images, which are randomly allocated into training and testing sets at a 4:1 ratio for the experiments.

TABLE I
Detail of the mAP Metrics

metric	detail
mAP	mAP at IoU = 0.50 : 0.05 : 0.95
$mAP@.50$	mAP at IoU = 0.50
$mAP@.75$	mAP at IoU=0.75
$mAP@_S$	mAP for small objects: area $<32^2$
$mAP@_M$	mAP for medium objects: 32^2 <area $<96^2$
$mAP@_L$	mAP for large objects: area $>96^2$

TABLE II
Detail of the mAR Metrics

metric	detail
mAR	mAR for all objects
$mAR@_S$	mAR for small objects: area $<32^2$
$mAR@_M$	mAR for medium objects: 32^2 <area $<96^2$
$mAR@_L$	mAR for large objects: area $>96^2$

According to the widely used criteria [43], we classify the objects into small, medium, and large ones based on their absolute area. Among them, small objects are those with a resolution of less than 32 $\times$ 32, medium objects are those with a resolution of less than 96 $\times$ 96 and greater than 32 $\times$ 32, and large objects have a resolution of greater than 96 $\times$ 96. The size distributions of objects for the two datasets are shown in Fig. 10.

B. Evaluation Metrics

In order to concisely assess and contrast the results of various detectors, two key metrics are employed: mean average precision (mAP) and mean average recall (mAR).

1) Mean Average Precision: The mA P is a metric used to evaluate the precision of detector, which can be subdivided into six types as shown in Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE COMPARISONS ON THE VISDRONE DATASET

Models	mAP	mAP@.50	mAP@.75	$mAP@_S$	$mAP@_M$	$mAP@_L$	mAR	$mAR@_S$	$mAR@_M$	$mAR@_L$
SSD300 [28]	0.110	0.218	0.102	0.035	0.182	0.305	0.193	0.100	0.299	0.434
FreeAnchor [44]	0.200	0.336	0.209	0.114	0.300	0.367	0.294	0.193	0.423	0.525
GuidedAnchoring [45]	0.205	0.355	0.207	0.117	0.312	0.354	0.294	0.198	0.425	0.451
FoveaBox [46]	0.203	0.349	0.206	0.117	0.310	0.379	0.299	0.205	0.433	0.474
ATSS [47]	0.215	0.360	0.218	0.126	0.325	**0.405**	0.316	0.218	0.451	**0.541**
VFNet [48]	0.229	**0.383**	0.233	0.140	0.336	0.403	0.314	0.223	0.439	0.495
FSANet [49]	0.131	0.242	0.131	0.063	0.197	0.234	0.246	0.155	0.352	0.480
CCOD [50]	0.209	0.358	0.211	0.123	0.313	0.351	0.300	0.208	0.432	0.464
Cheng et al. [51]	0.169	0.307	0.169	0.080	0.267	0.339	0.272	0.170	0.413	0.497
RFLA [52]	0.123	0.235	0.120	0.050	0.198	0.285	0.301	0.199	0.442	0.535
FPSOD	**0.231**	0.380	**0.241**	**0.151**	**0.342**	0.368	**0.333**	**0.242**	**0.474**	0.462

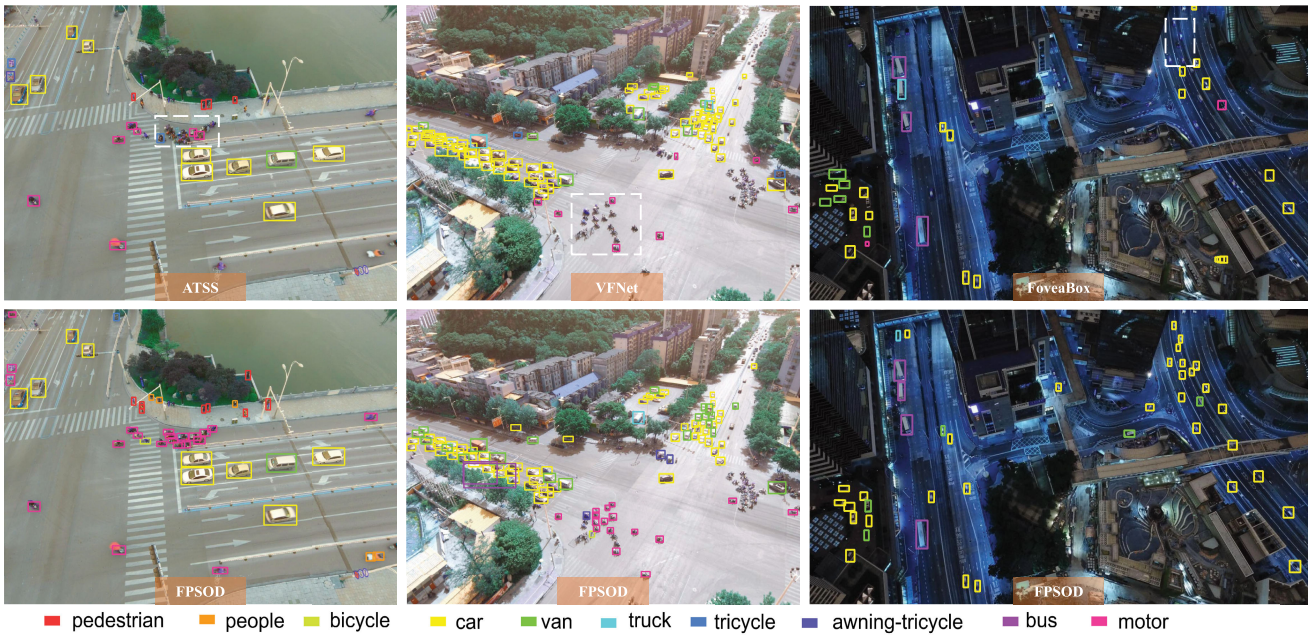


Fig. 11. Visualization of the predictions on the Visdrone dataset, with the missed detection indicated by the white dashed box.

2) Mean Average Recall: The mAR is a metric used to measure detector recall. To measure recall for all objects, including small, medium, and large objects, we utilize four refinement metrics: mAR, mAR@$_S$, mAR@$_M$, and mAR@$_L$, which are presented in Table II.

C. Implementation Details

ResNet-50 is utilized as the feature extraction backbone network in our model. In the proposed ASTFM, the convolutional kernel is set to 5, and the values of σ and n utilized in PR-Loss are 3/2 and 2, respectively. At each point on the multiscale feature map within the detection head, only one anchor is positioned with a scale of 1:1. Additionally, the stochastic gradient descent optimizer is utilized in this study, with the momentum of 0.90 and weight decay of 0.0001. The model is trained with a batch size of 4 for 100 epochs, with an initial learning rate of 0.001 that is reduced by a factor of 10 at the 60th and 90th epoch.

D. Comparison Results

To demonstrate the effectiveness of the FPSOD, we compare it against several object detection models including SSD [28], FreeAnchor [44], GuidedAnchoring [45], FoveaBox [46], ATSS [47], and VFNet [48]. In addition, this analysis includes recent algorithms that are relevant for small object detection. These algorithms are FSANet [49], CCOD [50], Cheng et al. [51], and RFLA [52].

For a fair comparison, we maintain consistent network settings, including the ResNet-50 backbone and optimization options, for both the compared models and FPSOD, except for the SSD model, which utilizes the VGG16 network from the original paper. The hyperparameters of the compared models are kept the same as those in the original papers, and data augmentation strategies of image resizing and random flipping are adopted.

1) Results on the Visdrone Dataset: Table III presents the quantitative results on the Visdrone dataset, showing a significant improvement of our model in most of the metrics. Specifically, our proposed FPSOD model outperforms all the compared algorithms in terms of mAP, indicating its superior detection accuracy. Moreover, the model exhibits the best performance in the mAP@$_s$ metric, thereby demonstrating its superior capability in detecting small objects. Additionally, FPSOD outperforms other models in recall metrics, achieving decent levels in all metrics.

Fig. 11 illustrates the visualization predictions of the compared models and the FPSOD on the Visdrone dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE COMPARISONS ON THE NWPU VHR-10 DATASET

Models	mAP	mAP@.50	mAP@.75	$mAP@_S$	$mAP@_M$	$mAP@_L$	mAR	$mAR@_S$	$mAR@_M$	$mAR@_L$
SSD300 [28]	0.484	0.870	0.499	0.151	0.425	0.592	0.554	0.256	0.491	0.653
FreeAnchor [44]	0.550	0.895	0.609	0.320	0.495	0.632	0.619	0.325	0.558	0.680
GuidedAnchoring [45]	0.541	0.910	0.600	0.233	0.501	0.614	0.613	0.271	0.565	0.664
FoveaBox [46]	0.542	0.888	0.602	0.258	0.497	0.630	0.608	0.267	0.560	0.679
ATSS [47]	0.555	0.927	0.614	0.253	0.508	0.618	0.642	0.290	0.588	0.680
VFNet [48]	0.560	0.907	0.641	0.135	0.521	0.607	0.634	0.296	0.584	0.679
FSANet [49]	0.391	0.811	0.349	0.277	0.388	0.377	0.582	0.328	0.567	0.549
CCOD [50]	0.528	0.870	0.591	0.277	0.505	0.589	0.602	0.277	0.583	0.623
Cheng et al. [51]	0.496	0.849	0.548	0.239	0.449	0.574	0.565	0.238	0.551	0.601
RFLA [52]	0.411	0.790	0.381	0.233	0.370	0.507	0.496	0.233	0.470	0.555
FPSOD	**0.577**	**0.928**	**0.649**	**0.344**	**0.532**	**0.638**	**0.659**	**0.373**	**0.605**	**0.700**

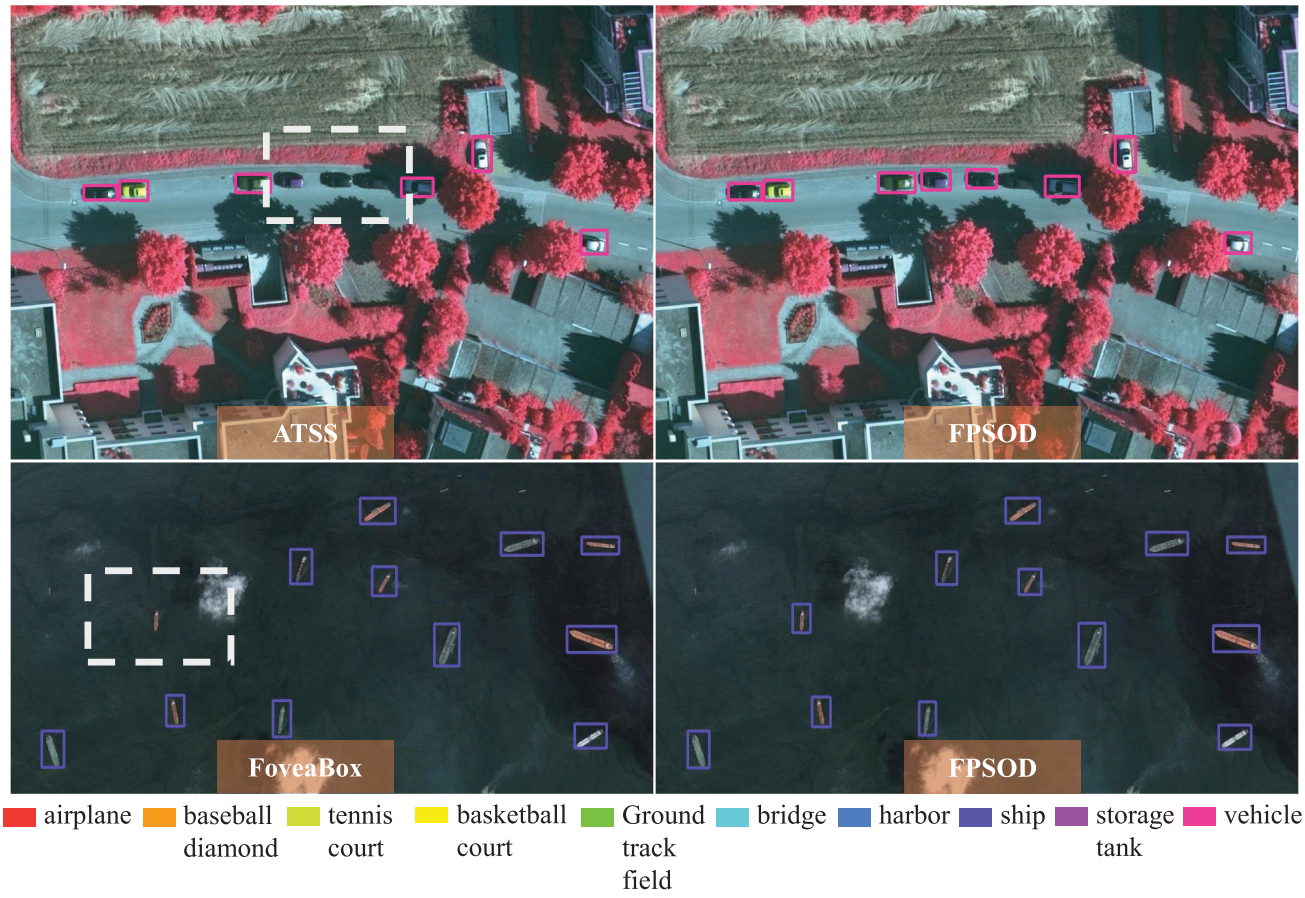


Fig. 12. Visualization of the predictions on the NWPU VHR-10 dataset, with the missed detection indicated by the white dashed box.

In contrast to the compared models, the proposed FPSOD demonstrates exceptional capability in detecting small objects, as evidenced by its successful identification of the person riding the motor in the first two columns. Additionally, it should be noted that RSIs may be captured under low-light conditions, as depicted in the right column. This can lead to a reduction in the clarity of information pertaining to the objects' features. The present findings indicate that the FPSOD remains capable of detecting a majority of objects, thereby substantiating its generalizability.

2) Results on the NWPU VHR-10 Dataset: Table IV displays the quantitative results, demonstrating that the FPSOD outperforms the compared models across all metrics, while also preserving the capacity for detecting large objects and mitigating the risk of missing small objects. These results provide evidence for the high precision detect ability of the proposed model across multiple scales.

The visualization results of the models on the NWPU VHR-10 dataset are shown in Fig. 12. The results demonstrate that the FPSOD is able to detect small objects with high precision, even under challenging scenarios where small objects may be occluded, shadowed by other objects or have similarities to the background. These issues commonly interfere with the feature representation of small objects, thus hindering the detection of small objects in RSI.

E. Ablation Experiments

1) Effect of PR-Loss and ASTFM: To assess the effectiveness of the proposed modules, three variants are performed: removing ASTFM (Ablation1), replacing PR-GIoULoss with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EFFECTIVENESS OF ASTFM AND PR-LOSS ON THE VISDRONE DATASET

Options	models			
use ASTFM ?		✓		✓
use PR-GIoULoss ?			✓	✓
Visdrone test mAP	0.215	0.221	0.224	**0.231**
Visdrone test $mAP@_S$	0.126	0.140	0.132	**0.151**

TABLE VI
EFFECTIVENESS OF ASTFM AND PR-LOSS ON THE NWPU VHR-10 DATASET

Options	models			
use ASTFM ?		✓		✓
use PR-GIoULoss ?			✓	✓
Visdrone test mAP	0.555	0.571	0.570	**0.577**
Visdrone test $mAP@_S$	0.253	0.325	0.285	**0.344**

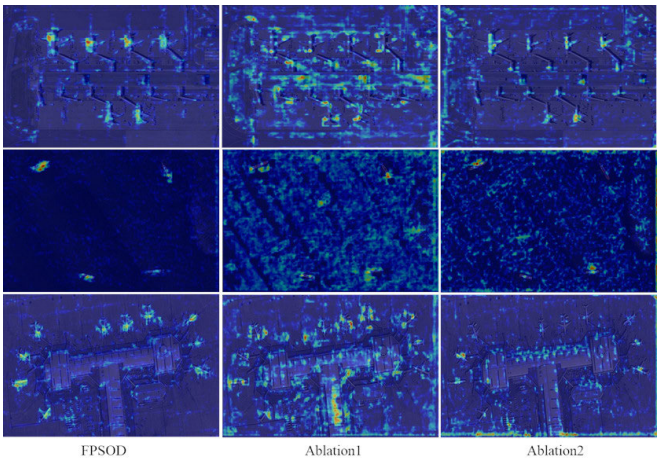


Fig. 13. Visualizations of ablations on the NWPU VHR-10 dataset.

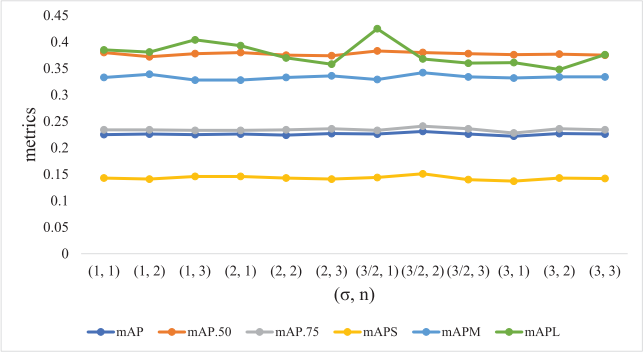


Fig. 14. Comparison of different σ and n on the Visdrone Dataset.

GIoULoss (Ablation2), and performing both operations simultaneously (Ablation3). In the ablation experiments, we focus on the precision for all objects (mAR) and for small objects (mAR@$_S$). The results in Tables V and VI indicate that both modules contribute to the precision of the proposed model. The incorporation of the ASTFM module brings about improvements of 0.6%–1.6% in the mAR metrics, as well as 1.4%–7.2% in the mAR@$_S$ metrics, for both datasets, respectively. Furthermore, the utilization of PR-GIoULoss yields improvements of 0.9%–0.6% in the mAR and mAR@$_S$ measurements for the Visdrone dataset, and 1.5%–3.2% for the NWPU VHR-10 dataset. When two modules are integrated, the performance of the model is optimized, indicating a

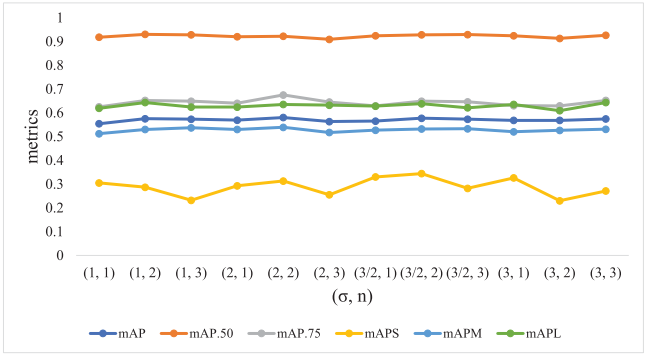


Fig. 15. Comparison of different σ and n on the NWPU VHR-10 Dataset.

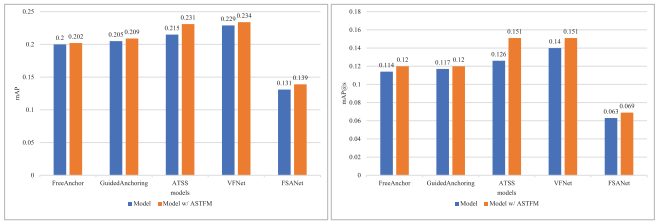


Fig. 16. Effectiveness of using ASTFM on other models on the Visdrone dataset.

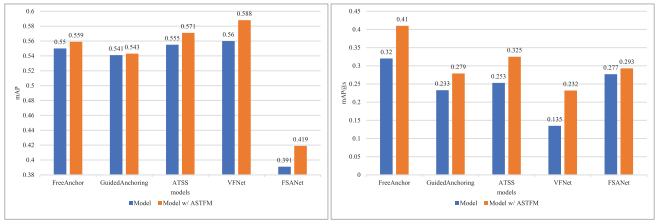


Fig. 17. Effectiveness of using ASTFM on other models on the NWPU VHR-10 dataset.

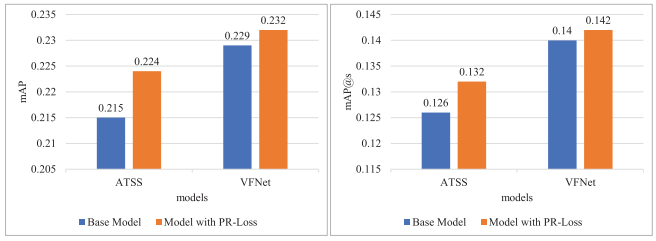


Fig. 18. Effectiveness of using PR-Loss on other models on the Visdrone dataset.

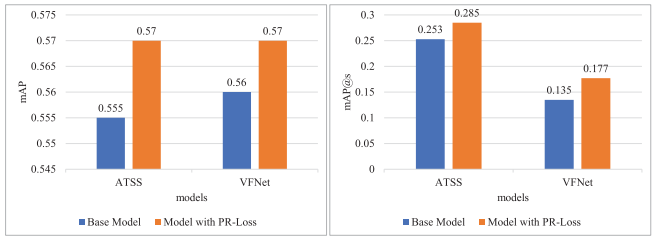
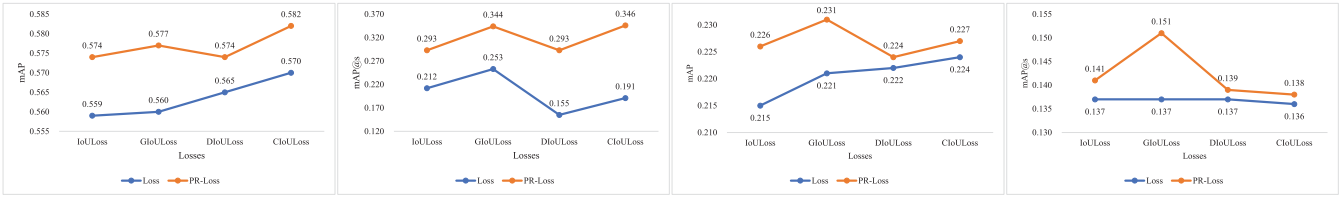


Fig. 19. Effectiveness of using PR-Loss on other models on the NWPU VHR-10 dataset.

notable positive impact of both modules and confirming the effectiveness of the progressive mechanism.

To visually demonstrate the contributions of the proposed PR-Loss and ASTFM, we present the visual results of Ablation1 and Ablation2 in Fig. 13. It can be observed that the attention maps in the first column (FPSOD) exhibit a distinct concentration of attention on the objects, characterized by the absence of extraneous feature interference. Upon the removal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 20. Comparison of the metrics from different loss functions on both datasets. (a) Results on the Visdrone dataset. (b) Results on the NWPU VHR-10 dataset.

TABLE VII
PERFORMANCE VARIATIONS WITH DIFFERENT k VALUES FOR THE VISDRONE DATASET

k	mAP	mAP@.50	mAP@.75	$mAP@_S$	$mAP@_M$	$mAP@_L$
k=3	0.228	0.374	0.238	0.141	0.341	0.353
k=5	**0.231**	**0.380**	**0.241**	**0.151**	0.342	0.368
k=7	0.230	**0.380**	**0.241**	0.146	**0.344**	**0.382**

TABLE VIII
PERFORMANCE VARIATIONS WITH DIFFERENT k VALUES FOR THE NWPU VHR-10 DATASET

k	mAP	mAP@.50	mAP@.75	$mAP@_S$	$mAP@_M$	$mAP@_L$
k=3	0.570	0.921	**0.650**	0.311	0.524	0.624
k=5	**0.577**	**0.928**	0.649	**0.344**	**0.532**	**0.638**
k=7	0.570	0.925	0.647	0.326	0.531	0.621

of the ASTFM module (Ablation1), a considerable presence of chaotic features is discernible within the attention maps. Furthermore, these features are situated in locations unrelated to the object under detection, consequently leading to an indeterminate determination of the object localization. Through a comparative analysis of the attention maps from the first column (FPSOD) and the third column (Ablation2), it becomes evident that the elimination of the PR-Loss component leads to a notable decrease in the attention directed toward the object in the attention maps of the third column. Furthermore, the attention region inadequately encompasses the complete extent of the object, thus providing incomplete coverage. The visual results presented in Fig. 13 offer compelling evidence of the efficacy of ASTFM in mitigating the presence of interfering features during the upsampling process. Additionally, the incorporation of PR-Loss demonstrates its capacity to enhance the network's proficiency in accurately localizing the objects.

2) Convolution Kernel Parameter k in ASTFM: In our experiments, we alter the convolution kernel parameter k within the ASTFM module, selecting classical convolution kernel sizes of 3, 5, and 7. Tables VII and VIII present results obtained from the Visdrone and NWPU VHR-10 datasets, respectively. The observations indicate that the model achieves optimal performance at $k = 5$. Nevertheless, the performance of the model is generally robust and exhibits insignificant variation with respect to the value of k.

3) σ and n in PR-Loss: The experimental results of varying the values of σ and n in PR-Loss on both datasets are displayed in Figs. 14 and 15. The figures indicates that just one metric per dataset is sensitive to variations in the σ and n values. Specifically, the metrics that are affected are denoted as mAR@$_L$ and mAR@$_S$. The differing results between the performances of σ and n on the two datasets may be attributed to variations in the quantity of objects and the conditions under which the data is obtained. The optimal balance between all metrics is attained by assigning values of 3/2 and 2 to σ and n, respectively.

4) Using ASTFM and PR-Loss on Other Models: Both the ASTFM and PR-Loss modules proposed are plug-and-play. To verify their generalizability, we apply them to other models and evaluate their performances using the mAR and mAR@$_S$.

1) *For ASTFM:* We incorporate the ASTFM module into the FPN-based models for comparison, and the results are presented in Figs. 16 and 17 for the two datasets. The results suggest that integrating ASTFM can enhance the effectiveness of models on both datasets, especially in terms of the mAR@$_S$.
2) *For PR-Loss:* To verify the effectiveness of PR-Loss, two experiments are conducted as follows.
   a) *Discuss 1:* Adding the proposed PR-Loss to the compared models. We replace the GIoULoss function in ATSS and VFNet with PR-GIoULoss. Other models are not evaluated as they utilized loss functions based on IoU. Figs. 18 and 19 illustrate the results obtained from the Visdrone and NWPU VHR-10 datasets, respectively.
   b) *Discuss 2:* Generalization of the PR-Loss form to other loss functions. We train the FPSOD using various loss functions along with their PR-formed counterparts. The comparison results are depicted in Fig. 20.

The experimental results above indicate that PR-Loss can not only be incorporated into different models to improve their performance, but can also be applied to various forms of loss functions, demonstrating excellent versatility.

V. CONCLUSION

In this article, we propose a one-stage RSI object detection model based on the progressive mechanism, which is particularly effective at enhancing the performance of detecting small objects. The model initially builds a soft-threshold filtering-based feature pyramid structure, which filters redundant features during the upsampling process and improves the semantic richness of small objects. Additionally, we propose a PR-Loss function that can adaptively weight gradients for different IoUs and can be generalized to multiple forms. The loss function can reduce the possibility of prediction bounding box drift of small objects and thus improve their localization accuracy. We evaluate the performance of the proposed model on the Visdrone and NWPU VHR-10 datasets, achieving 15.1%–34.4% mAR@$_s$, respectively, outperforming all compared models. Furthermore, we conduct ablation studies to demonstrate the effectiveness of our proposed modules.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

In future work, we aim to further improve the ability of the detector to accurately detect small objects.

REFERENCES

[1] M. B. Bejiga, A. Zeggada, and F. Melgani, "Convolutional neural networks for near real-time object detection from UAV imagery in avalanche search and rescue operations," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 693–696.

[2] K. Chen, "Military object detection using multiple information extracted from hyperspectral imagery," in *Proc. Int. Conf. Prog. Inform. Comput. (PIC)*, Dec. 2017, pp. 124–128.

[3] V. Muthukkumarasamy, M. Blumenstein, J. Jo, and S. Green, "Intelligent illicit object detection system for enhanced aviation security," in *Proc. Int. Conf. Simulated Evol. Learn.*, 2004, pp. 1–6.

[4] N. M. Noor, A. Abdullah, and M. Hashim, "Remote sensing UAV/drones and its applications for urban areas: A review," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 169, Jul. 2018, Art. no. 012003.

[5] A. Bannari, D. Morin, G. B. Bénié, and F. J. Bonn, "A theoretical review of different mathematical models of geometric corrections applied to remote sensing images," *Remote Sens. Rev.*, vol. 13, nos. 1–2, pp. 27–47, Aug. 1995.

[6] Z. Wang, R. Wang, J. Ai, H. Zou, and J. Li, "Global and local context-aware ship detector for high-resolution SAR images," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 4, pp. 4159–4167, Aug. 2023.

[7] J. Cui, H. Jia, H. Wang, and F. Xu, "A fast threshold neural network for ship detection in large-scene SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6016–6032, 2022.

[8] G. X. Hu, Z. Yang, L. Hu, L. Huang, and J. M. Han, "Small object detection with multiscale features," *Int. J. Digit. Multimedia Broadcast.*, vol. 2018, pp. 1–10, Sep. 2018.

[9] J. Leng, Y. Ren, W. Jiang, X. Sun, and Y. Wang, "Realize your surroundings: Exploiting context information for small object detection," *Neurocomputing*, vol. 433, pp. 287–299, Apr. 2021.

[10] C. Chen, W. Gong, Y. Chen, and W. Li, "Object detection in remote sensing images based on a scene-contextual feature pyramid network," *Remote Sens.*, vol. 11, no. 3, p. 339, Feb. 2019.

[11] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[12] J.-S. Lim, M. Astrid, H.-J. Yoon, and S.-I. Lee, "Small object detection using context and attention," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIIC)*, 2021, pp. 181–186.

[13] B. Huo, C. Li, J. Zhang, Y. Xue, and Z. Lin, "SAFF-SSD: Self-attention combined feature fusion-based SSD for small object detection in remote sensing," *Remote Sens.*, vol. 15, no. 12, p. 3027, Jun. 2023.

[14] H. Liang, Y. Yang, Q. Zhang, L. Feng, J. Ren, and Q. Liang, "Transformed dynamic feature pyramid for small object detection," *IEEE Access*, vol. 9, pp. 134649–134659, 2021.

[15] H.-J. Park, J.-W. Kang, and B.-G. Kim, "ssFPN: Scale sequence (S^2) feature-based feature pyramid network for object detection," *Sensors*, vol. 23, no. 9, p. 4432, Apr. 2023.

[16] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.

[17] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13668–13677.

[18] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.

[19] X. Wang, D. Zhu, and Y. Yan, "Towards efficient detection for small objects via attention-guided detection network and data augmentation," *Sensors*, vol. 22, no. 19, p. 7663, Oct. 2022.

[20] F. Xiaolin et al., "Small object detection in remote sensing images based on super-resolution," *Pattern Recognit. Lett.*, vol. 153, pp. 107–112, Jan. 2022.

[21] S. M. A. Bashir and Y. Wang, "Small object detection in remote sensing images with residual feature aggregation-based super-resolution and object detector network," *Remote Sens.*, vol. 13, no. 9, p. 1854, May 2021.

[22] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.

[23] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[24] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[25] S. Qiao, L.-C. Chen, and A. Yuille, "DetectoRS: Detecting objects with recursive feature pyramid and switchable atrous convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10213–10224.

[26] Z. Liu et al., "RelationRS: Relationship representation network for object detection in aerial images," *Remote Sens.*, vol. 14, no. 8, p. 1862, Apr. 2022.

[27] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[28] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[29] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[30] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.

[31] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox: An advanced object detection network," in *Proc. 24th ACM Int. Conf. Multimedia*, Oct. 2016, pp. 516–520.

[32] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.

[33] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.

[34] Y.-F. Zhang, W. Ren, Z. Zhang, Z. Jia, L. Wang, and T. Tan, "Focal and efficient IOU loss for accurate bounding box regression," 2021, *arXiv:2101.08158*.

[35] J. He, S. Erfani, X. Ma, J. Bailey, Y. Chi, and X.-S. Hua, "α-IoU: A family of power intersection over union losses for bounding box regression," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 20230–20242.

[36] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[37] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[38] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[39] P. Zhu et al., "Detection and tracking meet drones challenge," 2020, *arXiv:2001.06303*.

[40] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[41] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[42] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[43] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1222–1230.

[44] X. Zhang, F. Wan, C. Liu, X. Ji, and Q. Ye, "Learning to match anchors for visual object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3096–3109, Jun. 2022.

[45] J. Wang, K. Chen, S. Yang, C. C. Loy, and D. Lin, "Region proposal by guided anchoring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2965–2974.

[46] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyound anchor-based object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 7389–7398, 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.

[47] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[48] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, "VarifocalNet: An IoU-aware dense object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8514–8523.

[49] J. Wu, Z. Pan, B. Lei, and Y. Hu, "FSANet: Feature-and-spatial-aligned network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630717.

[50] Y. Yang, B. Zang, N. Li, B. Zhao, B. Li, and Y. Lang, "Reducing false detections in aerial images by exploiting the context information and centroid relationship," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–14, 2022.

[51] G. Cheng, M. He, H. Hong, X. Yao, X. Qian, and L. Guo, "Guiding clean features for object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[52] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 526–543.



Yang Yang (Member, IEEE) received the Ph.D. degree from Tianjin University, Tianjin, China, in 2019.

He is currently an Assistant Professor with Tianjin University. His research interests include deep learning, pattern recognition, object detection, human activity recognition, and micro-Doppler radar.



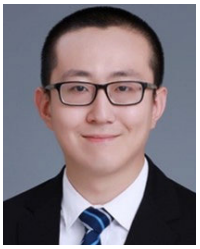
Bingjie Zang received the B.S. degree from the School of Computer and Communication Engineering, Northeastern University at Qinhuangdao, Qinhuangdao, China, in 2020. She is currently pursuing the M.A.Sc. degree with the School of Electrical and Information Engineering, Tianjin University, Tianjin, China.

Her research interests include deep learning and object detection.



Chunying Song received the B.Eng. degree from Tianjin University, Tianjin, China, in 2018, where she is currently pursuing the Ph.D. degree with the School of Electronic and Information Engineering.

Her research interests include deep learning, micro-Doppler radar, and image quality assessment.



Beichen Li received the B.E. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2010, and the M.S. degree from the Stevens Institute of Technology, Hoboken, NJ, USA, in 2016. He is currently pursuing the Ph.D. degree with Tianjin University, Tianjin, China, in the field of deep learning for computer vision and the Internet of Things.

He was a Research Assistant with the Maritime Security Center, SIT, from 2014 to 2015. He worked for several companies, including Good Technology and Kingsoft Cloud, Beijing, China. His working experiences include quality assurance, software development, cloud service, data analysis, and big data processing.



Yue Lang (Member, IEEE) received the M.Sc. and Ph.D. degrees from Tianjin University, Tianjin, China, in 2021.

She is currently a Lecturer with Hebei University of Technology, Tianjin. She was also a Visiting Student with Nanyang Technological University, Singapore, from 2020 to 2021. Her research interests include micro-Doppler effect, radar target recognition, and machine learning.



Wenyuan Zhang received the Ph.D. degree in geodesy and surveying engineering from China University of Mining and Technology, Xuzhou, China, in December 2022.

He is currently an Assistant Professor with China University of Mining and Technology. His research interests include GNSS meteorology and tropospheric tomography.



Peng Huo received the M.S. degree from Xi'an Jiaotong University, Xi'an, China, in 2015.

He is currently working with the High Voltage System Department, Longyuan (Beijing) Renewable Energy Engineering Technology Company Ltd., Beijing, China. His main research interests include reliability improvement of power transmission and transformation equipment in renewable energy station.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:31 UTC from IEEE Xplore. Restrictions apply.